# Memory Forensics Framework MFF v2

## Cross-Case Comparison Report

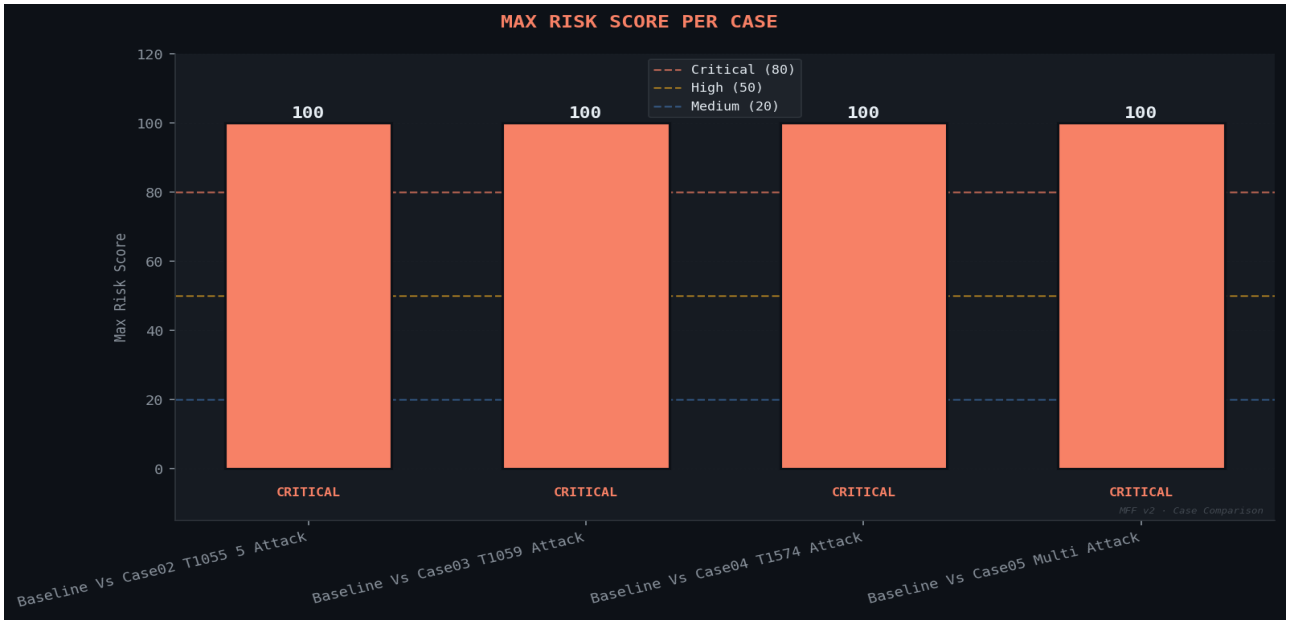
| | |
|---|---|
| **Generated** | 2026-04-19 18:52:15 UTC |
| **Cases compared** | 4 |
| **Case labels** | Baseline Vs Case02 T1055 5 Attack · Baseline Vs Case03 T1059 Attack · Baseline Vs Case04 T157 |

# 1. EXECUTIVE SUMMARY

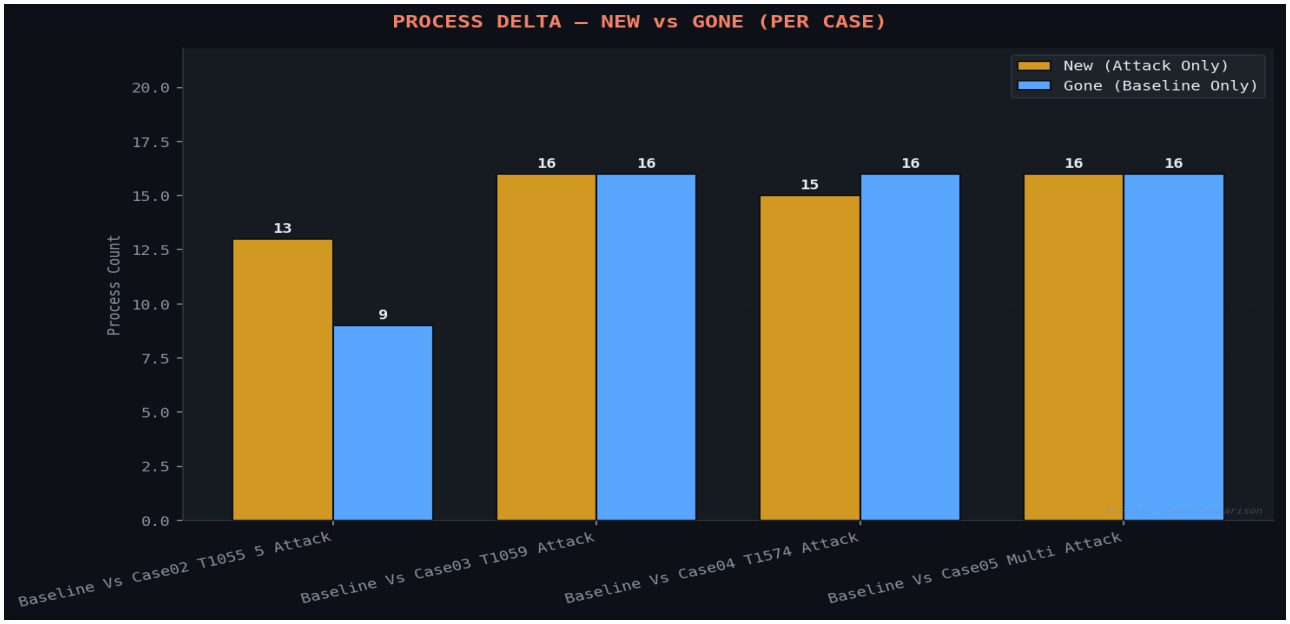
| Case | Severity | Max Score | Critical | Techniques | RWX | DLL Hijack | IOCs | New Procs |
|---|---|---|---|---|---|---|---|---|
| Baseline Vs Case02 T1055 5 Attack | CRITICAL | 100 | 3 | 2 | 7 | 2 | 40 | 13 |
| Baseline Vs Case03 T1059 Attack | CRITICAL | 100 | 3 | 2 | 5 | 1 | 52 | 16 |
| Baseline Vs Case04 T1574 Attack | CRITICAL | 100 | 6 | 3 | 22 | 9 | 49 | 15 |
| Baseline Vs Case05 Multi Attack | CRITICAL | 100 | 8 | 8 | 42 | 12 | 59 | 16 |

# 2. COMPARISON DASHBOARD

## 3. RISK SCORE COMPARISON

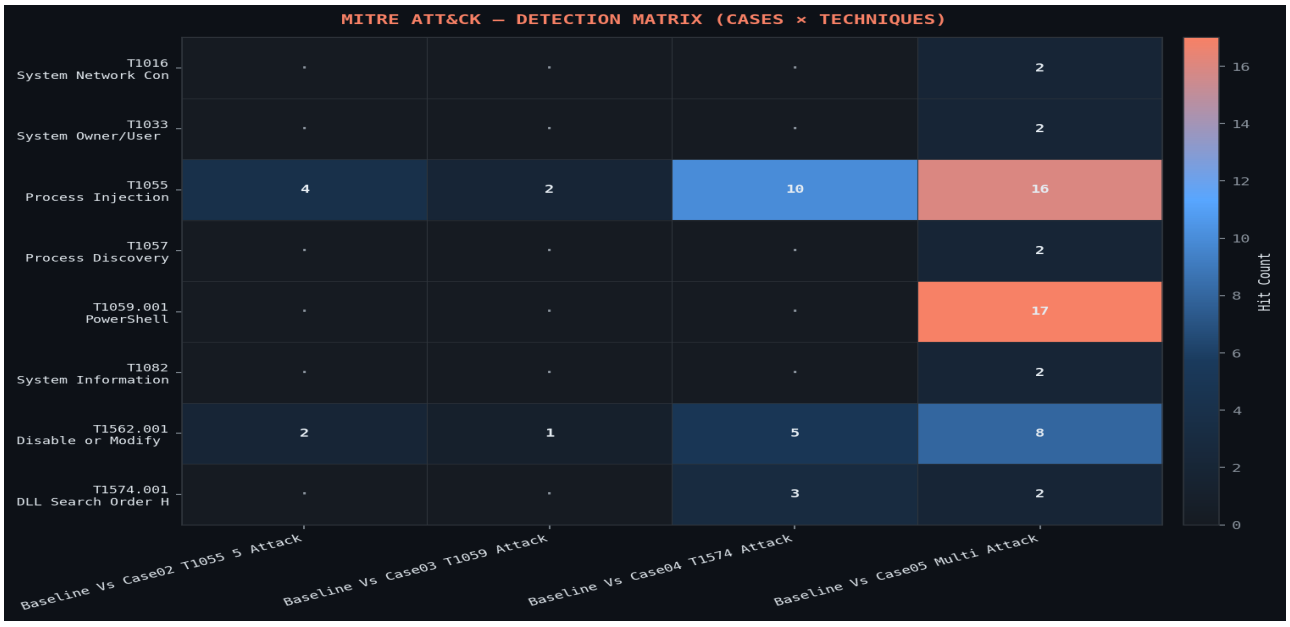



## 4. PROCESS DELTA

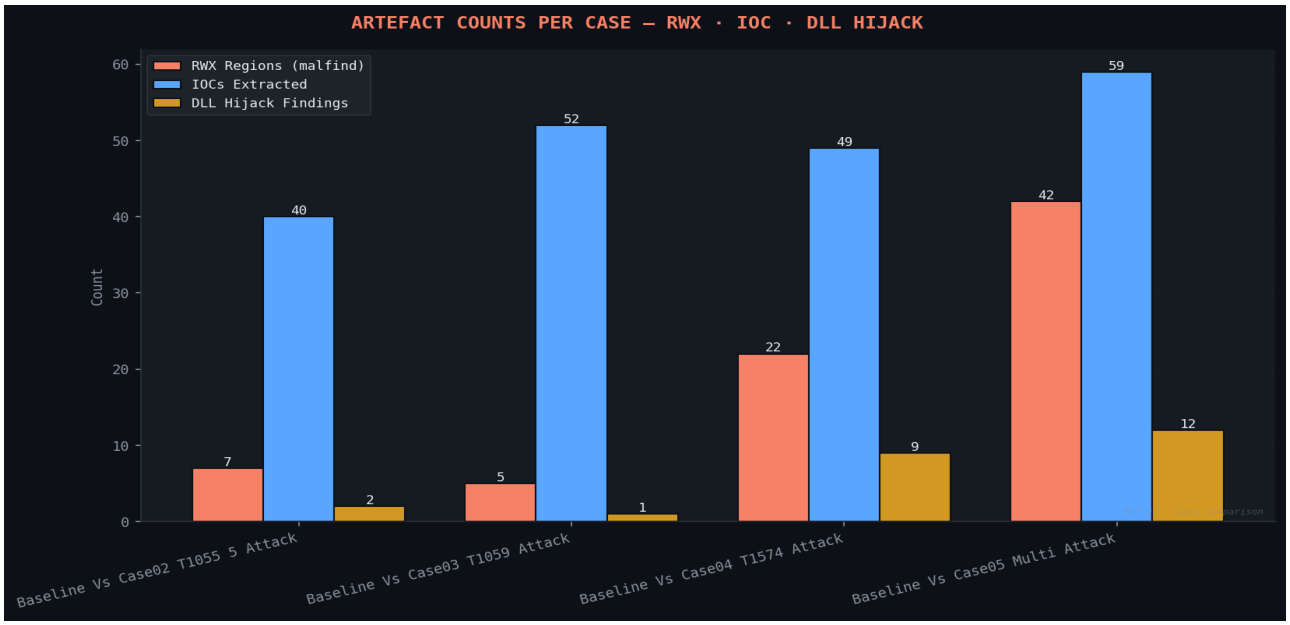



## 5. ATT&CK; TACTIC COVERAGE

## 6. MITRE ATT&CK; MATRIX




## 7. ARTEFACT COUNTS


ARTEFACT COUNTS PER CASE — RWX · IOC · DLL HIJACK
RWX Regions (malfind)
IOCs Extracted
DLL Hijack Findings
Count
0
10
20
30
40
50
60
7
40
2
5
52
1
22
49
9
42
59
12
Baseline Vs Case02 T1055 5 Attack
Baseline Vs Case03 T1059 Attack
Baseline Vs Case04 T1574 Attack
Baseline Vs Case05 Multi Attack

# 8. TECHNIQUE DETECTION MATRIX

| Technique | Tactic | Name | Baseline Vs Ca | Baseline Vs Ca | Baseline Vs Ca | Baseline Vs Ca |
|---|---|---|---|---|---|---|
| T1016 | Discovery | System Network Configuration Discovery | · | · | · | ✓ |
| T1033 | Discovery | System Owner/User Discovery | · | · | · | ✓ |
| T1055 | Defense Evasion | Process Injection | ✓ | ✓ | ✓ | ✓ |
| T1057 | Discovery | Process Discovery | · | · | · | ✓ |
| T1059.001 | Execution | PowerShell | · | · | · | ✓ |
| T1082 | Discovery | System Information Discovery | · | · | · | ✓ |
| T1562.001 | Defense Evasion | Disable or Modify Tools - AMSI Memory Patch | ✓ | ✓ | ✓ | ✓ |
| T1574.001 | Defense Evasion | DLL Search Order Hijacking | · | · | ✓ | ✓ |

# 9. INDIVIDUAL CASE SUMMARIES

## Baseline Vs Case02 T1055 5 Attack

| | |
|---|---|
| **Severity** | CRITICAL |
| **Highest risk process** | powershell.exe (score: 100) |
| **Techniques detected** | T1055, T1562.001 |
| **New processes** | 13 |
| **RWX memory regions** | 7 |
| **DLL hijack findings** | 2 |
| **IOCs extracted** | 40 |
| **Net new connections** | 131 |

## Baseline Vs Case03 T1059 Attack

| | |
|---|---|
| **Severity** | CRITICAL |
| **Highest risk process** | powershell.exe (score: 100) |
| **Techniques detected** | T1055, T1562.001 |
| **New processes** | 16 |
| **RWX memory regions** | 5 |
| **DLL hijack findings** | 1 |
| **IOCs extracted** | 52 |
| **Net new connections** | 133 |

## Baseline Vs Case04 T1574 Attack

| | |
|---|---|
| **Severity** | CRITICAL |
| **Highest risk process** | powershell.exe (score: 100) |
| **Techniques detected** | T1055, T1562.001, T1574.001 |
| **New processes** | 15 |
| **RWX memory regions** | 22 |
| **DLL hijack findings** | 9 |
| **IOCs extracted** | 49 |
| **Net new connections** | 94 |

## Baseline Vs Case05 Multi Attack

| | |
|---|---|
| **Severity** | CRITICAL |
| **Highest risk process** | powershell.exe (score: 100) |
| **Techniques detected** | T1016, T1033, T1055, T1057, T1059.001, T1082, T1562.001, T1574.001 |
| **New processes** | 16 |
| **RWX memory regions** | 42 |
| **DLL hijack findings** | 12 |
| **IOCs extracted** | 59 |
| **Net new connections** | 139 |

# 10. METHODOLOGY

**Comparison method:** Each attack case analysed independently against the same baseline using comparison_engine_v2.py. Results (CSV + JSON) loaded by this comparison module and aggregated into cross-case charts and tables.

**Risk scoring:** Rule-based transparent scoring (no ML). RWX +60 · Suspicious cmdline +40 · Malicious name +50 · DLL hijack +30 · Anomaly bonus +15.

**DLL analysis:** dll_analysis.py checks windows.dlllist CSV for protected DLLs (amsi.dll, version.dll, etc.) loaded from non-System32 paths.

**ATT&CK; mapping:** mitre_tagger.py — 30+ rule-based signatures across 8 tactics. Includes T1574.001 DLL hijacking, T1562 defense evasion, T1059 execution.

**Framework:** MFF v2 — Python Post-Volatility Memory Forensics Framework. FYP — University of Roehampton 2026. Modules: comparison_engine_v2, dll_analysis, mitre_tagger, network_ioc, process_tree, case_comparison, report_generator.